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(54) SOLENOID VALVE POPPET AFFIXMENT

(71) Applicant: **ASCO, L.P.**, Florham Park, NJ (US)

(72) Inventors: **James R. Ward**, Sturgis, MI (US); **Alexander R. Determan**, Grand Blanc, MI (US); **Matthew Tyler Hoskins**, Oak Park, MI (US); **Rakeshkumar Krishnat Patil**, Pune (IN); **Niles Hanmant Sapate**, Pune (IN)

(73) Assignee: **ASCO, L.P.**, Florham Park, NJ (US)

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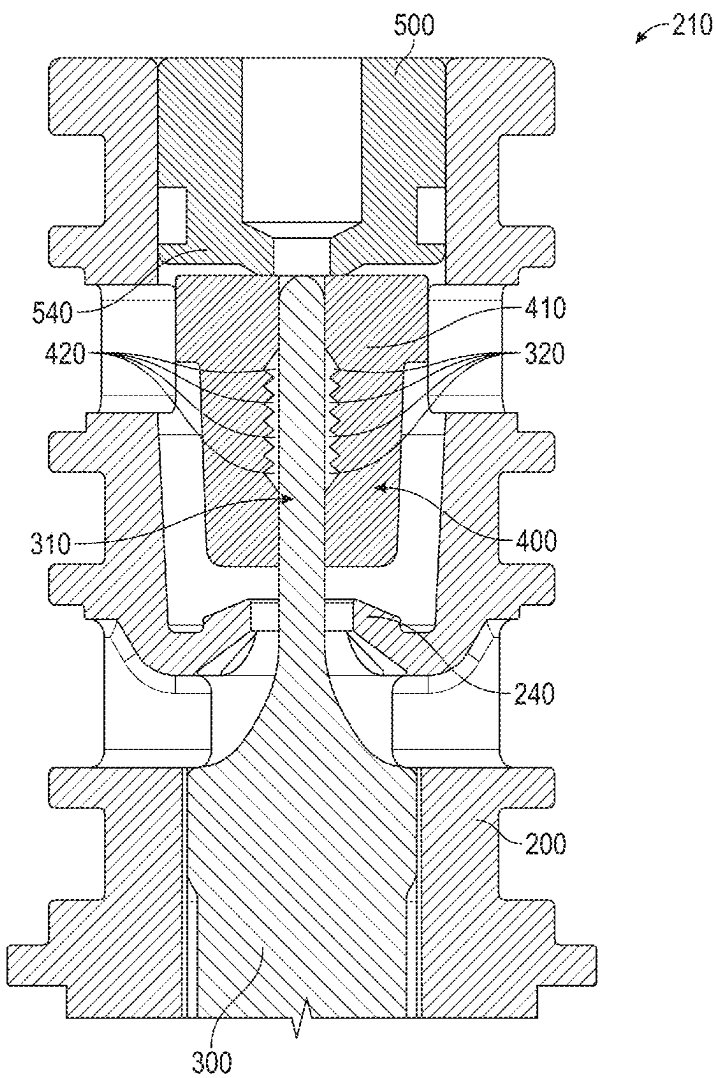
F16K 11/044	(2006.01)

(52) **U.S. Cl.**

CPC ***F16K 31/0627*** (2013.01); ***F16K 1/36*** (2013.01); ***F16K 1/42*** (2013.01); ***F16K 11/044*** (2013.01)

(57) **ABSTRACT**

A valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, and a poppet configured to selectively engage a valve seat to open and close a flow path through the port section of the valve body. The plunger can include a stem extending into the port section of the valve body. The stem can include a plurality of annular barbs. The barbs can engage an interior of the poppet to hold the poppet in place on the stem.



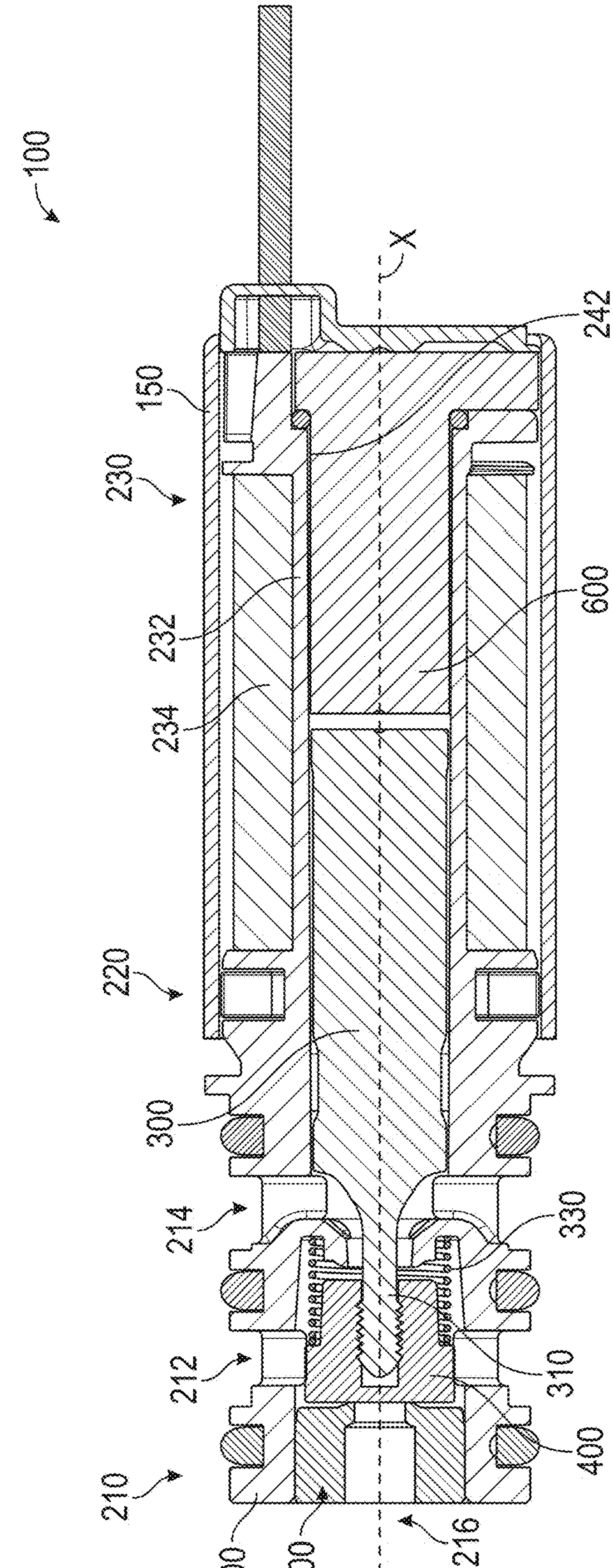

100
X
242
150
230
232
234
600
220
300
214
330
212
310
400
210
200
500
216

FIG. 1

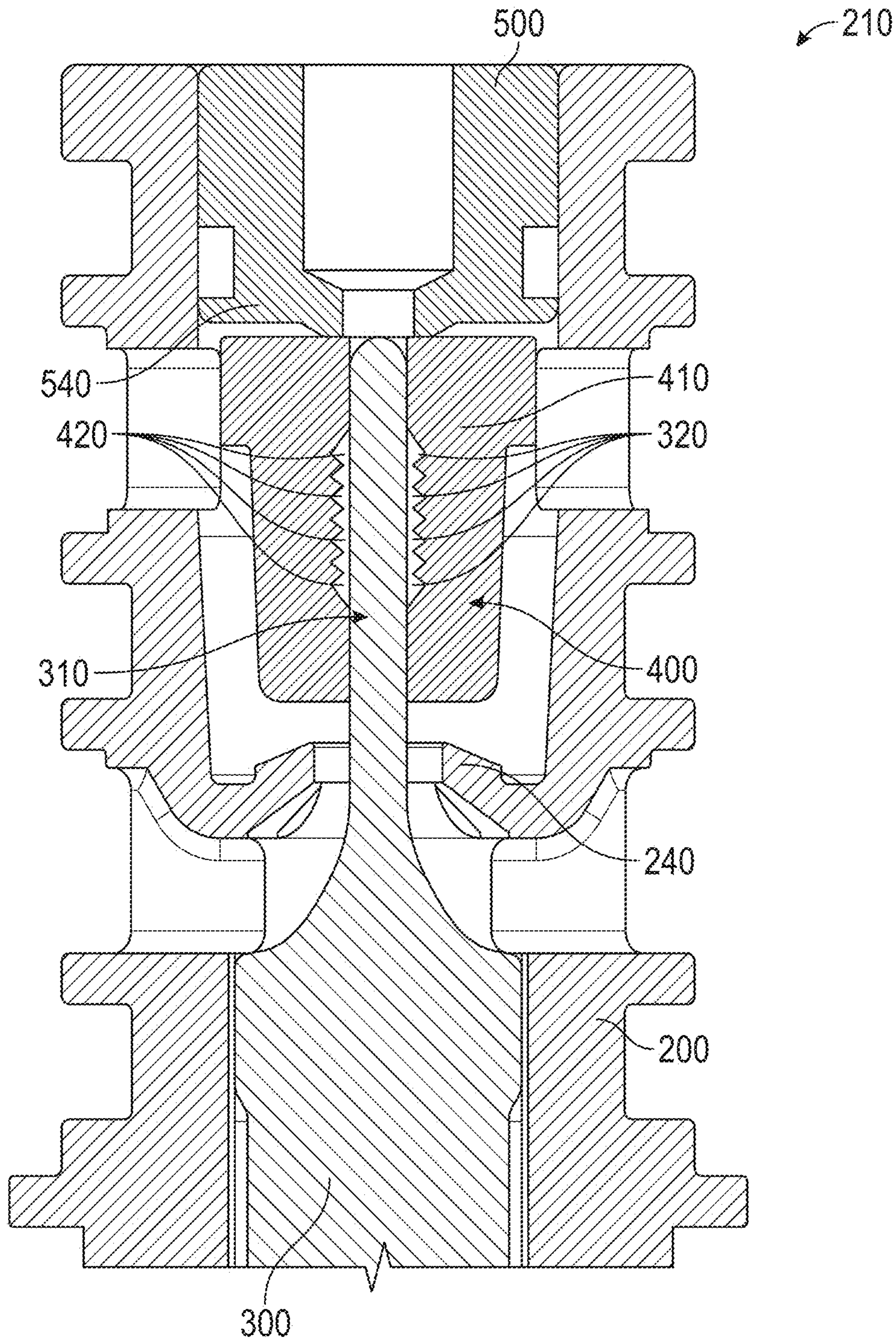


FIG. 2

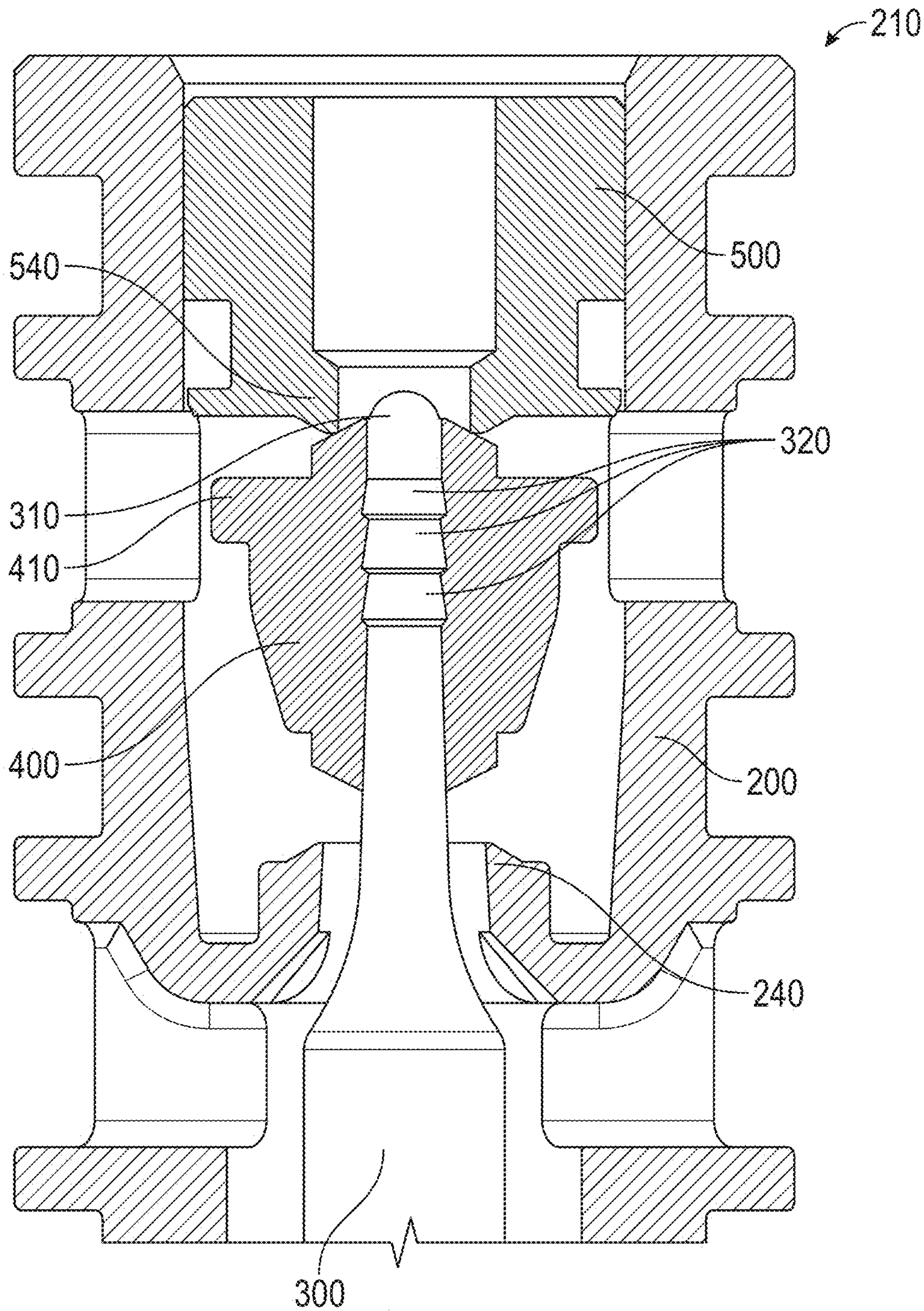


FIG. 3

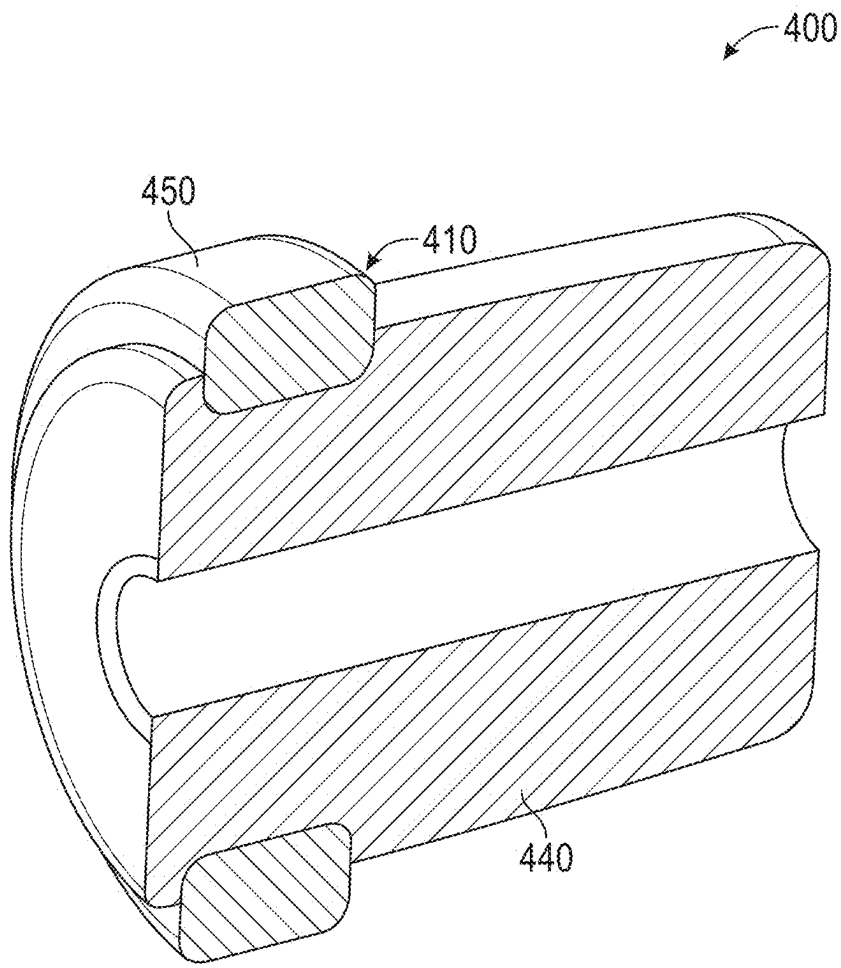


FIG. 4

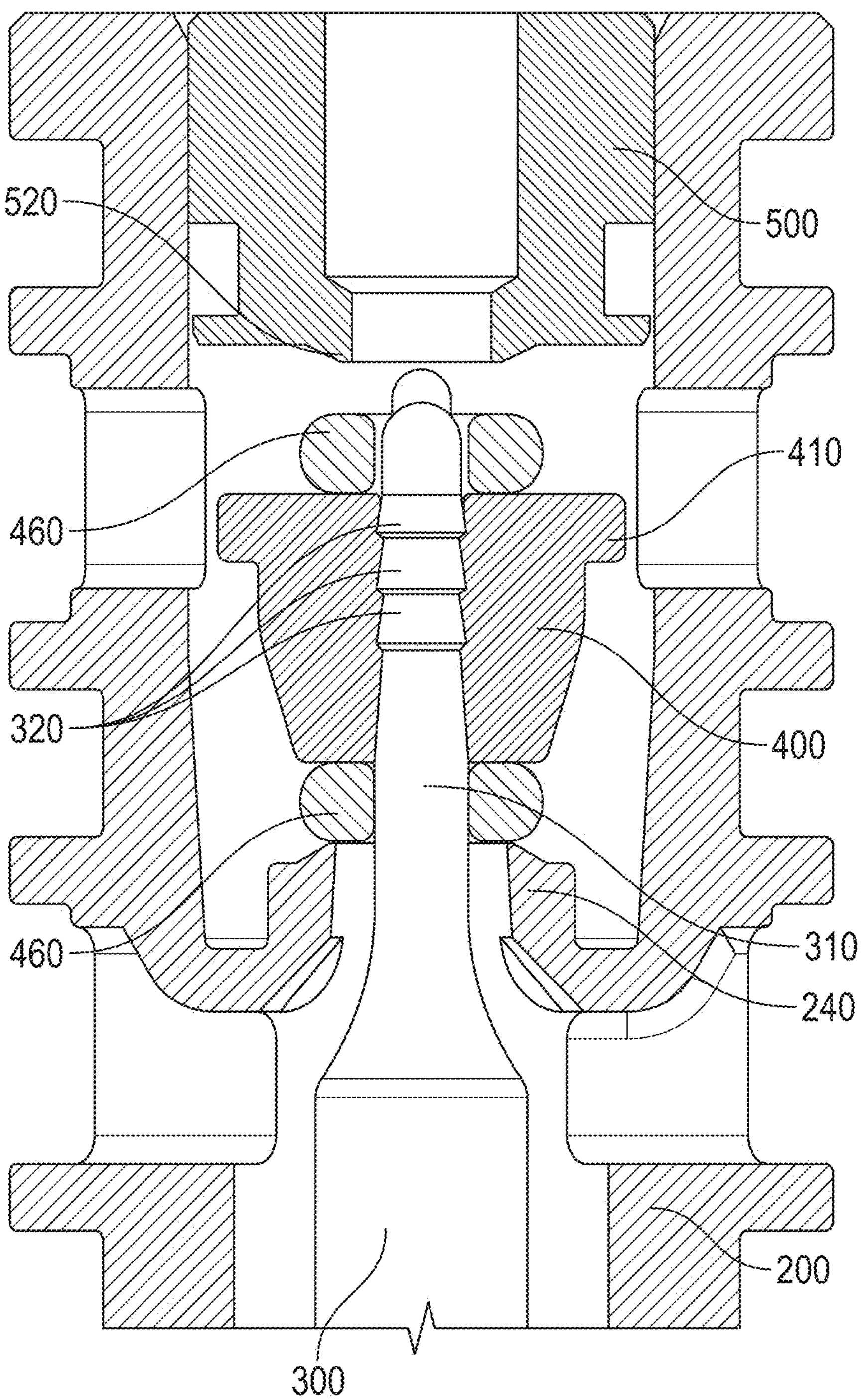


FIG. 5

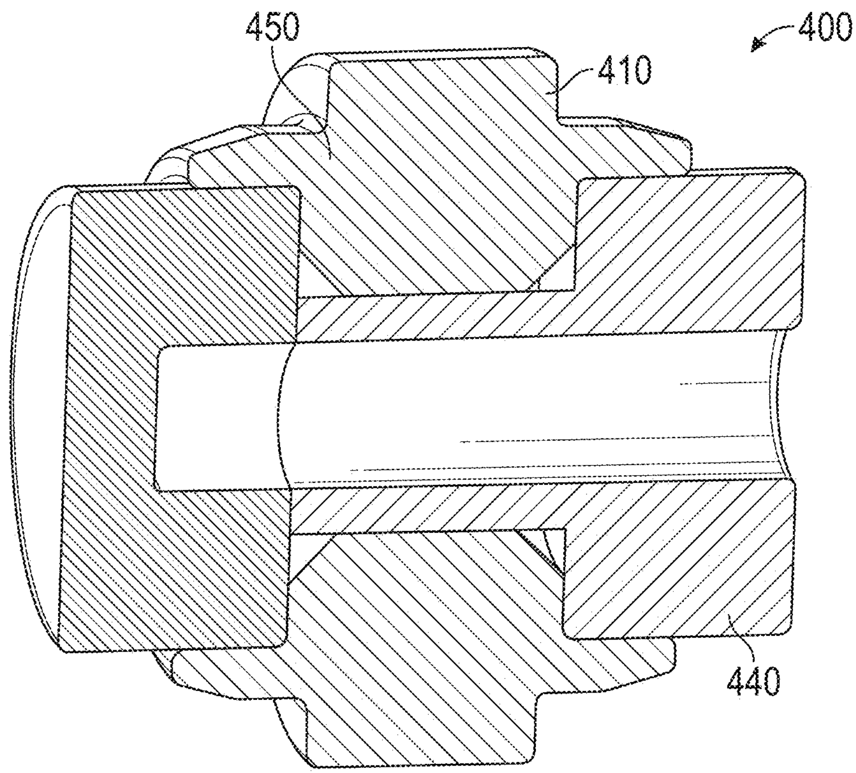


FIG. 6

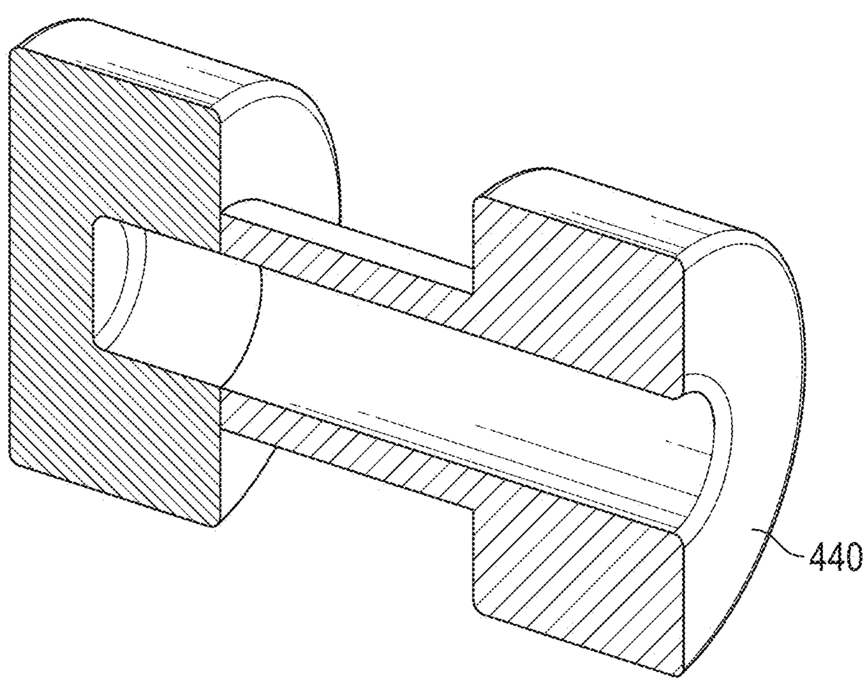


FIG. 7

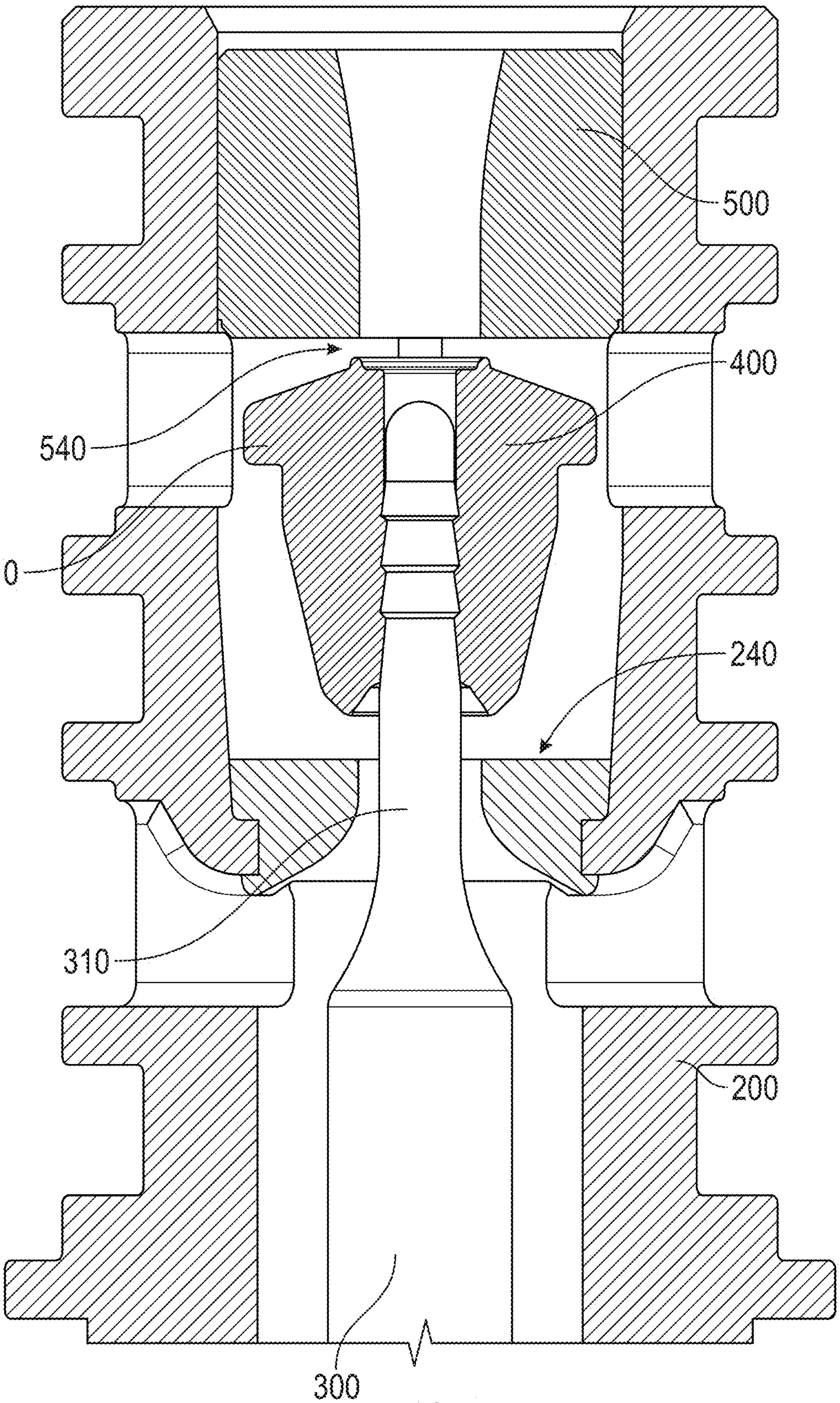


FIG. 8

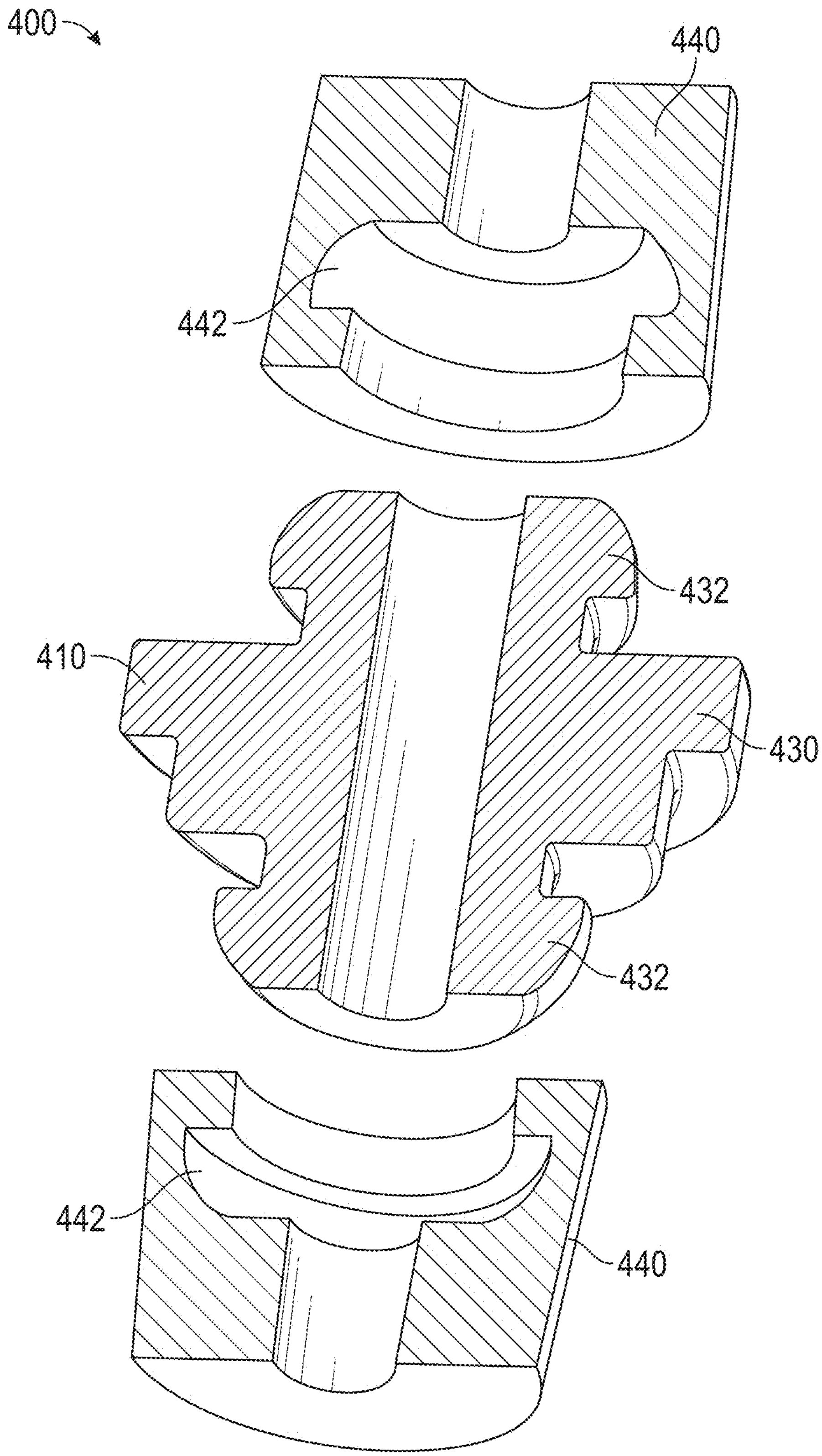


FIG. 9

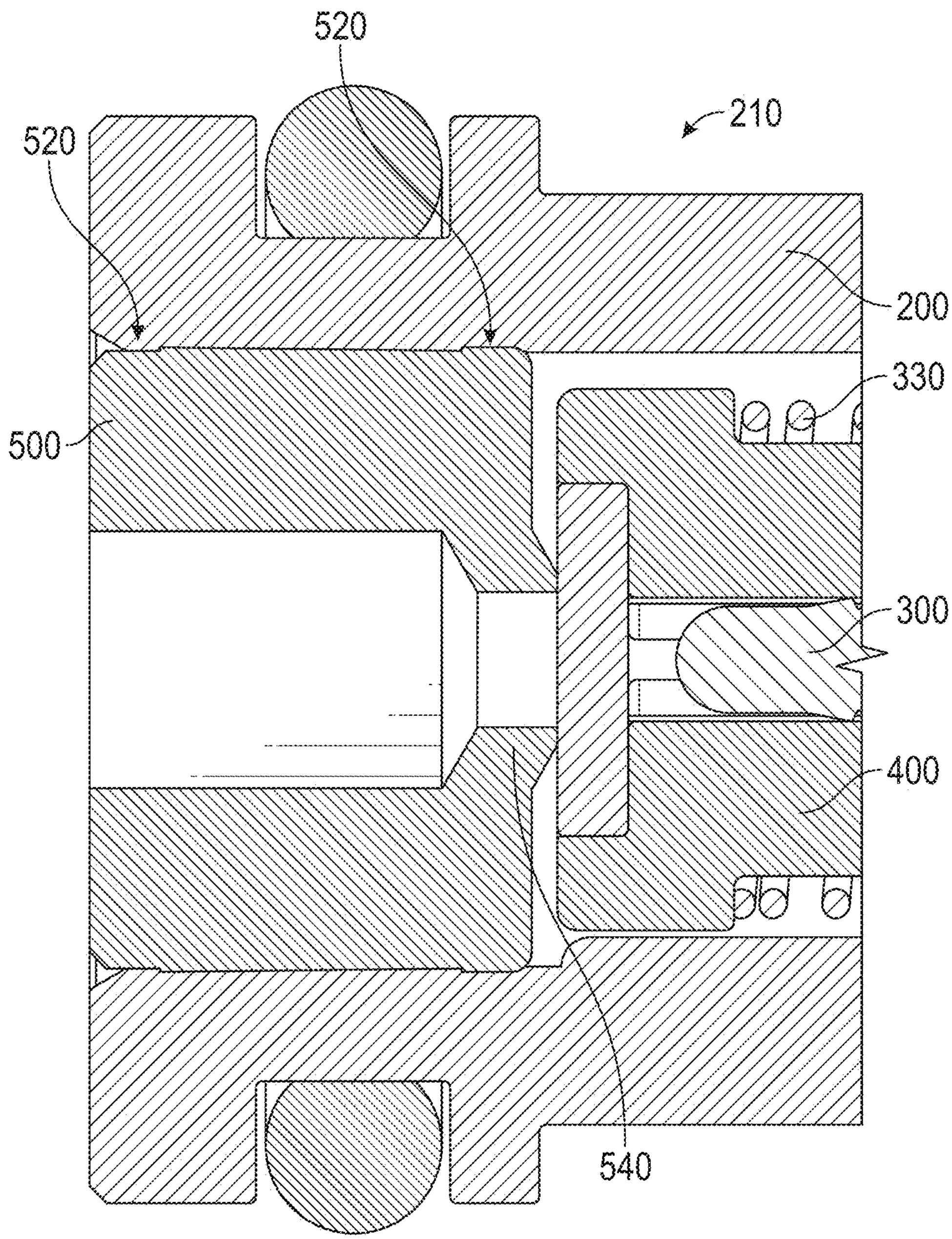


FIG. 10

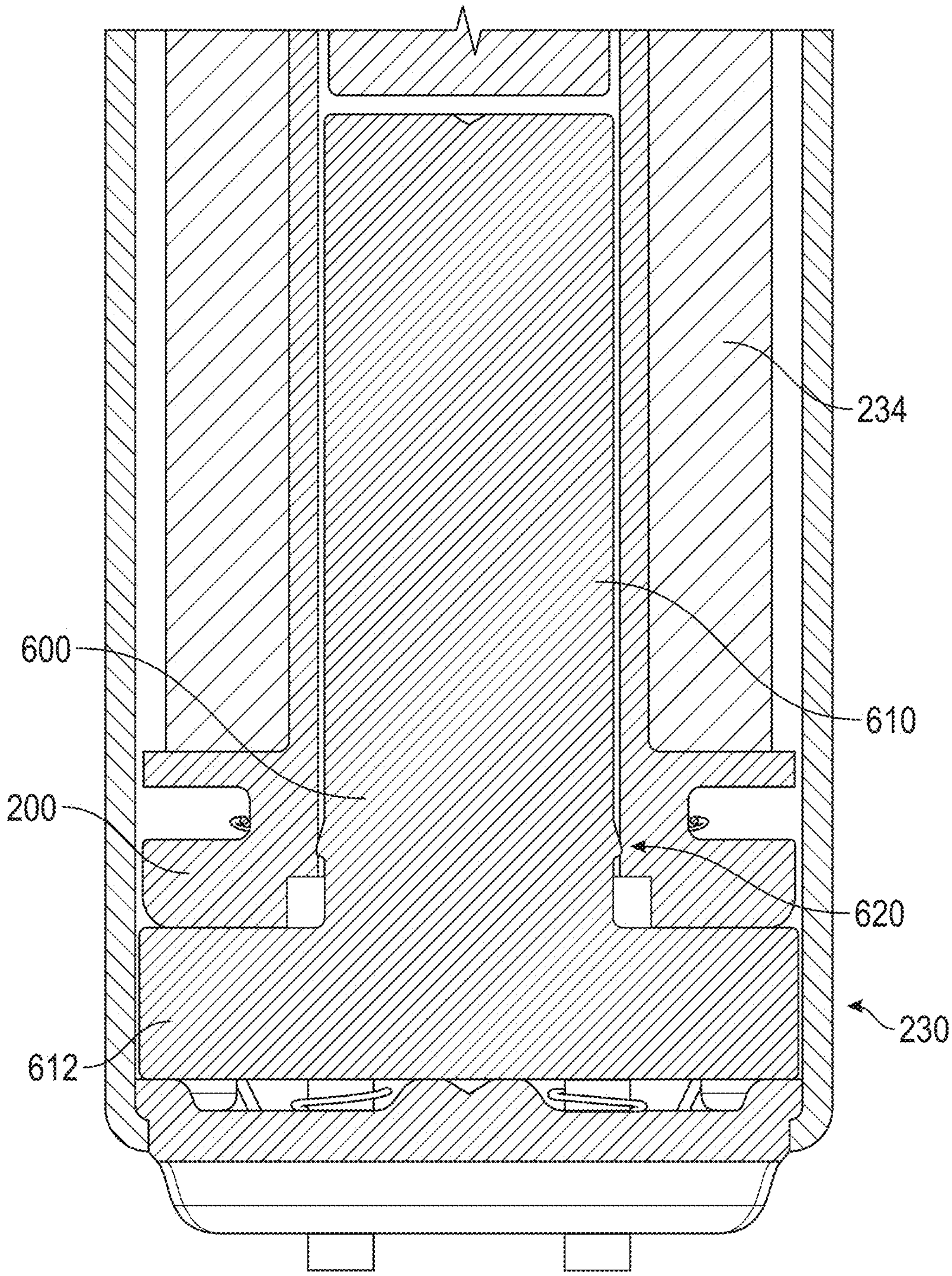


FIG. 11

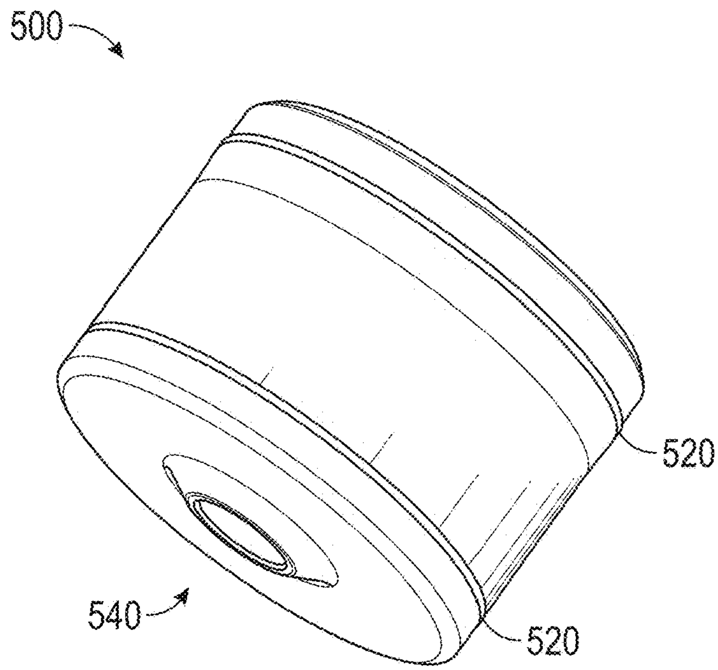


FIG. 12

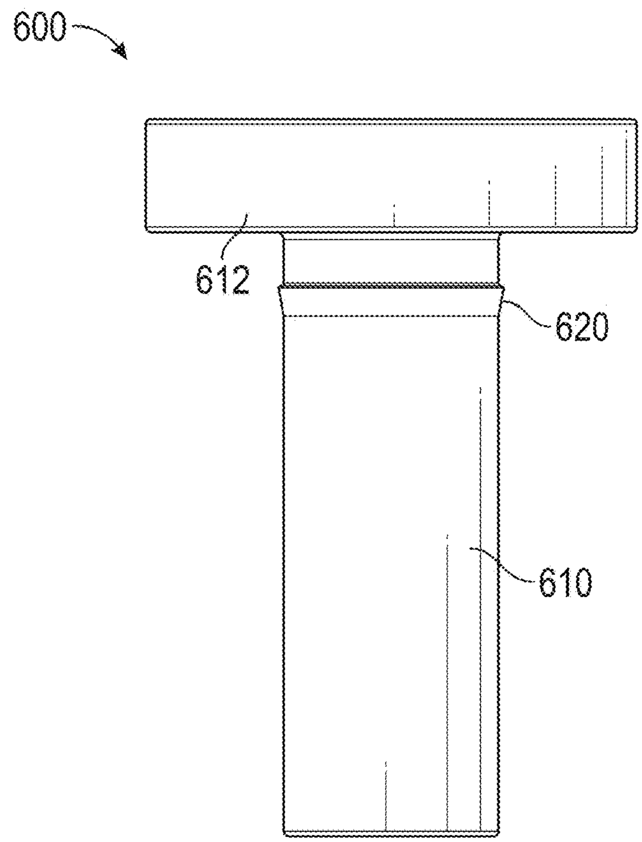


FIG. 13

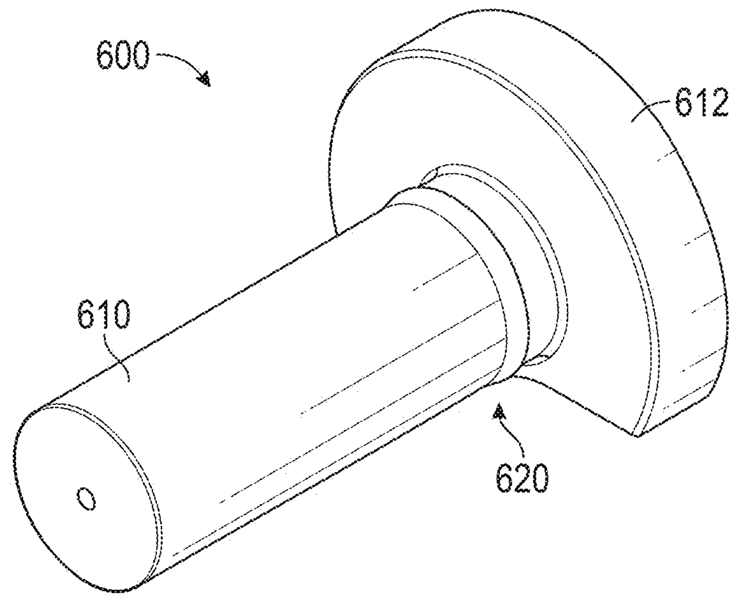


FIG. 14

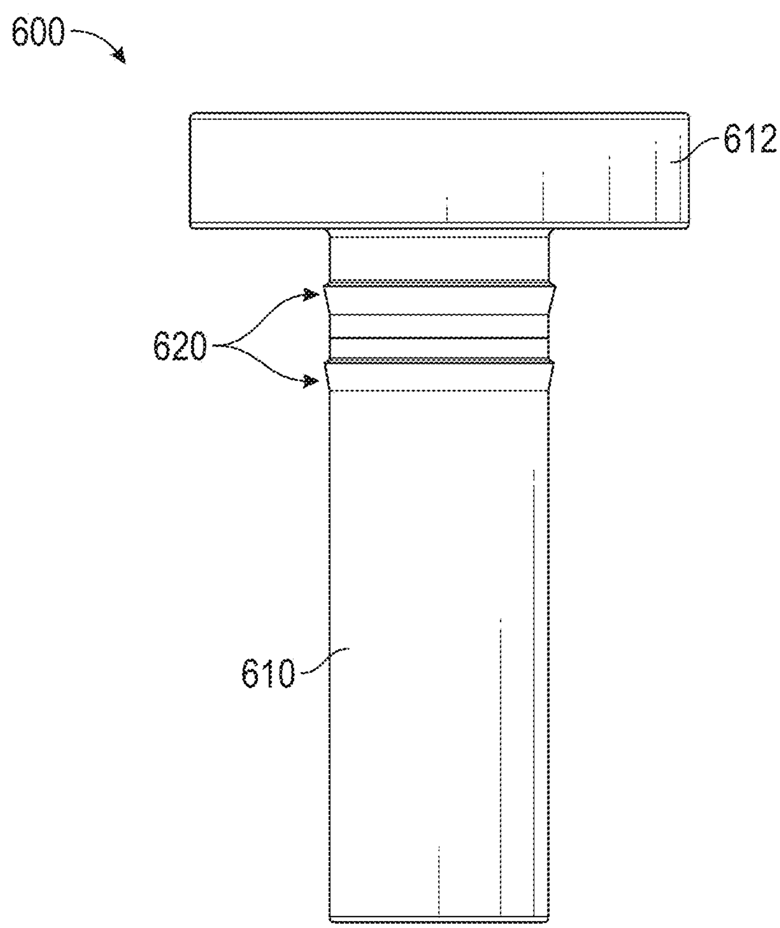


FIG. 15

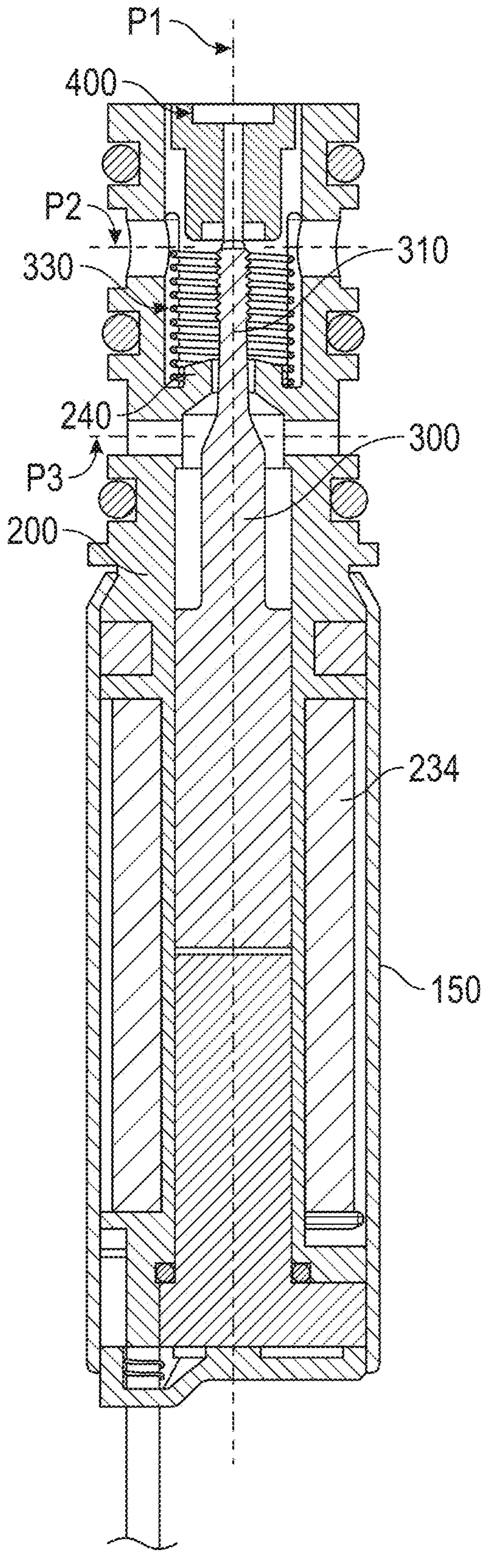


FIG. 16

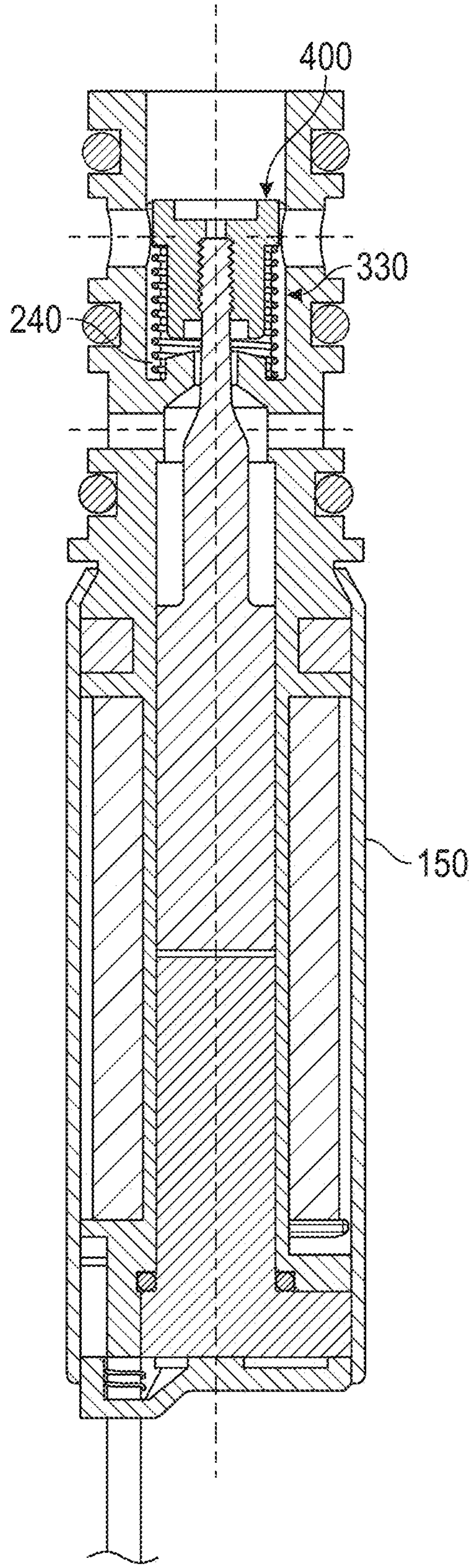


FIG. 17

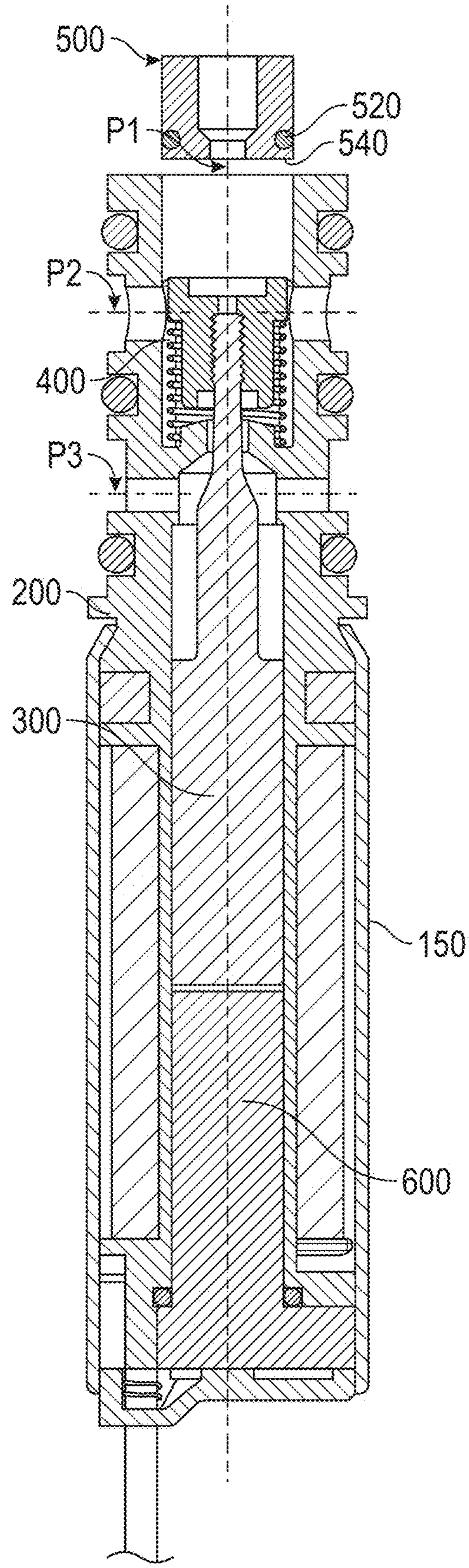


FIG. 18

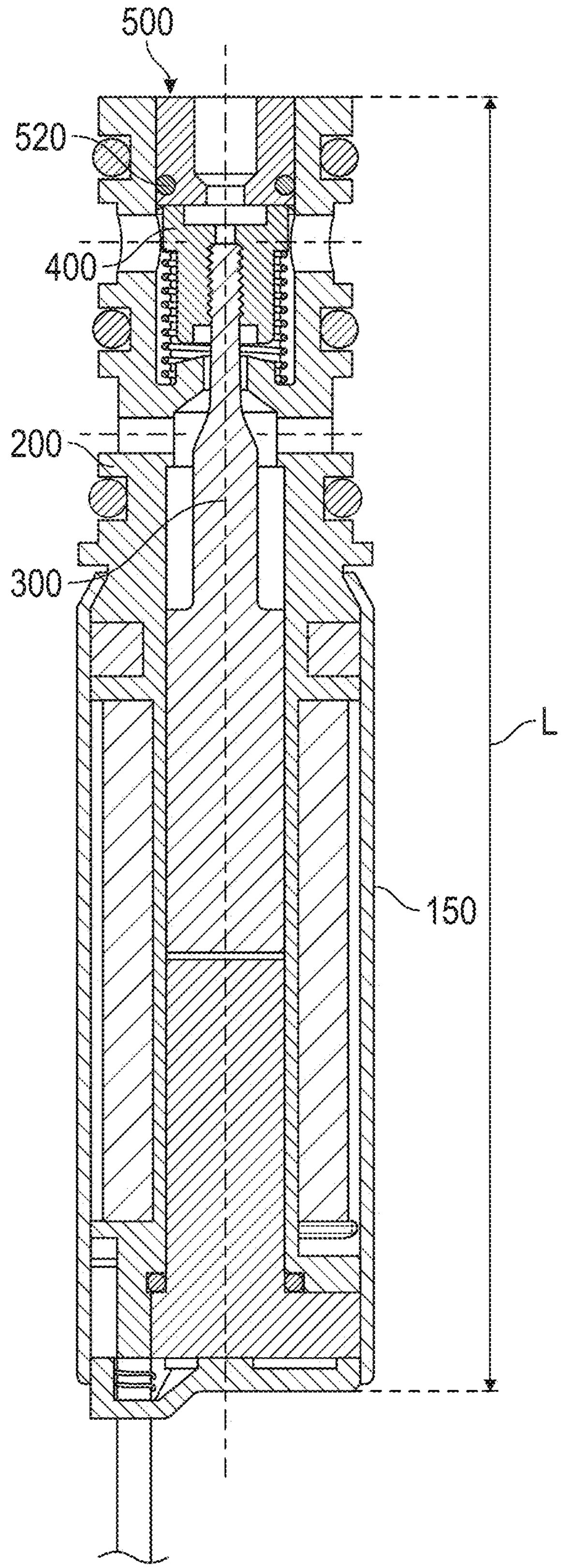


FIG. 19

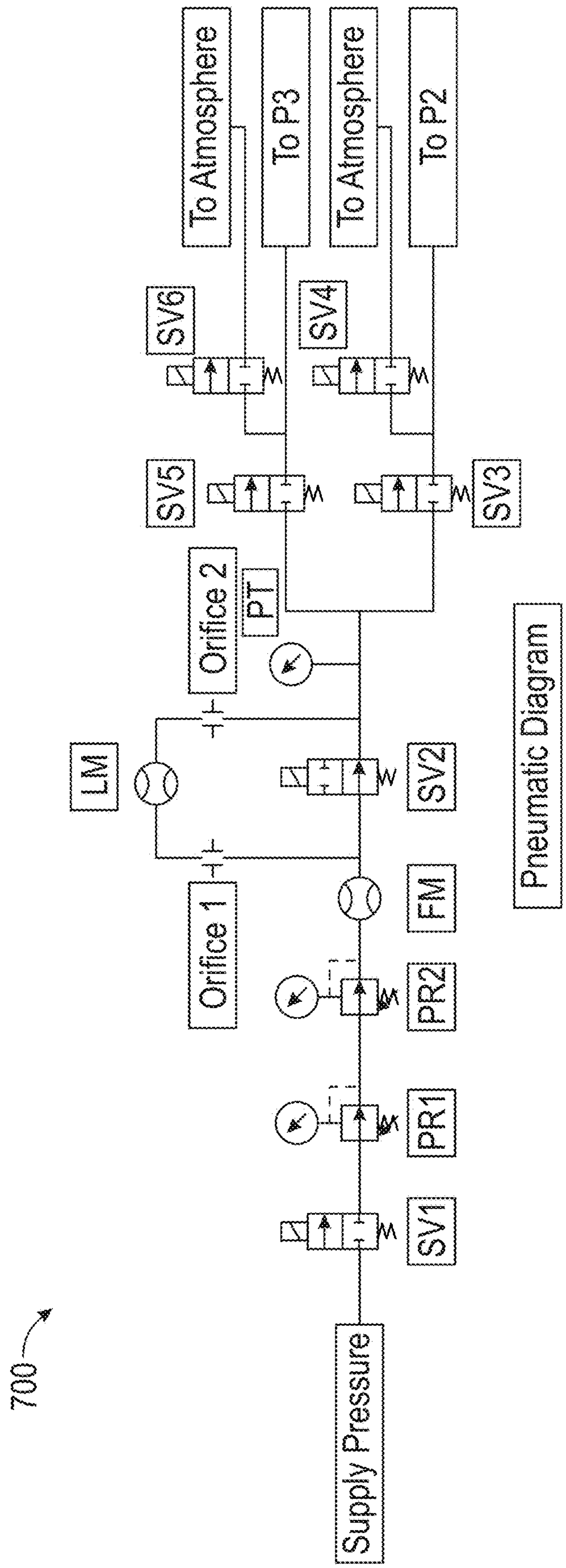

700
Supply Pressure
SV1
PR1
PR2
FM
Orifice 1
LM
SV2
Orifice 2
PT
SV5
SV3
SV6
SV4
To Atmosphere
To P3
To Atmosphere
To P2
Pneumatic Diagram

FIG. 20

SOLENOID VALVE POPPET AFFIXMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Indian Patent Application number 202421014854 filed Feb. 29, 2024, the contents of which are incorporated herein by reference. This application is related to commonly owned U.S. patent application Ser. Nos. 18/590,943, 18/590,944, 18/590,945 and 18/590,946, each filed on Feb. 29, 2024, and the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] Field of the Invention The present disclosure relates generally to solenoid valves and more specifically relates to miniature solenoid valves.

DESCRIPTION OF THE RELATED ART

[0003] Solenoid valves commonly have poppets to selectively engage valve seats to open and close flow paths therethrough. Currently, such poppets are often manufactured using complex over molding techniques, which can be tedious and time consuming, where alignment can be critical. Such problems often cause quality control issues. As solenoid valves get smaller, these issues are compounded.

SUMMARY OF THE INVENTION

[0004] Applicants have created new and useful devices, systems and methods for solenoid valves. In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet press-fit onto the plunger and configured to selectively open and close at least one flow path through the port section of the valve body, or any combination thereof. In at least one embodiment, the plunger can include a stem extending into the port section of the valve body. In at least one embodiment, the poppet can be press-fit onto the stem. In at least one embodiment, the stem can include one or more annular barbs to engage an interior of the poppet to hold the poppet in place on the stem. In at least one embodiment, the poppet can be configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body.

[0005] In at least one embodiment, the poppet can be rigid and/or configured to selectively engage a resilient valve seat to selectively open and close the flow path through the port section of the valve body. In at least one embodiment, the poppet can include a rigid portion configured to engage the barb on the stem and/or a resilient portion configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body. In at least one embodiment, the rigid portion can be harder than the resilient portion. In at least one embodiment, the rigid portion can include an annular flange. In at least one embodiment, the resilient portion can mate with the flange.

[0006] In at least one embodiment, the poppet can include a band around a perimeter of the poppet to secure the poppet to the stem. In at least one embodiment, the band can be crimped onto the poppet, thereby securing the poppet to the stem.

[0007] In at least one embodiment, the valve seat can be integral with the valve body. In at least one embodiment, the valve seat can be integral with a cap press-fit into one end of the valve body. In at least one embodiment, the poppet can include a seal configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body. In at least one embodiment, the poppet can include a first seal configured to selectively engage a first valve seat to selectively open and close a normally closed flow path through the port section of the valve body. In at least one embodiment, the poppet can include a second seal configured to selectively engage a second valve seat to selectively open and close a normally open flow path through the port section of the valve body.

[0008] In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet including a resilient portion configured to selectively engage a rigid valve seat to open and close at least one flow path through the port section of the valve body, or any combination thereof. In at least one embodiment, the plunger can include a stem extending into the port section of the valve body. In at least one embodiment, the stem can include at least one annular barb configured to engage an interior of the poppet to hold the poppet in place on the stem. In at least one embodiment, the valve seat can be integral with the valve body. In at least one embodiment, the valve seat can be integral with a cap press-fit into one end of the valve body.

[0009] In at least one embodiment, the poppet can include a rigid portion configured to engage the barb on the stem. In at least one embodiment, the rigid portion can be harder than the resilient portion. In at least one embodiment, the rigid portion can include an annular flange. In at least one embodiment, the resilient portion can mate with the flange.

[0010] In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet configured to selectively engage a valve seat to open and close a flow path through the port section of the valve body, or any combination thereof. In at least one embodiment, the plunger can include a stem extending into the port section of the valve body. In at least one embodiment, the stem can include a plurality of annular barbs. In at least one embodiment, the barbs can engage an interior of the poppet to hold the poppet in place on the stem.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a cross sectional view of one of many embodiments of a solenoid valve according to the disclosure.

[0012] FIG. **2** is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure.

[0013] FIG. **3** is a cross sectional view of a portion of another one of many embodiments of a solenoid valve according to the disclosure.

[0014] FIG. **4** is a cross sectional perspective view of one of many embodiments of a poppet according to the disclosure.

[0015] FIG. **5** is a cross sectional view of a portion of yet another one of many embodiments of a solenoid valve according to the disclosure.

[0016] FIG. **6** is a cross sectional perspective view of another one of many embodiments of a poppet according to the disclosure.

[0017] FIG. **7** is a cross sectional perspective view of a portion of the poppet of FIG. **6**.

[0018] FIG. **8** is a cross sectional view of a portion of still another of many embodiments of a solenoid valve according to the disclosure.

[0019] FIG. **9** is an exploded cross sectional view of a portion of one of many embodiments of a poppet according to the disclosure.

[0020] FIG. **10** is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure.

[0021] FIG. **11** is a cross sectional view of another portion of one of many embodiments of a solenoid valve according to the disclosure.

[0022] FIG. **12** is a perspective view of one of many embodiments of a cap according to the disclosure.

[0023] FIG. **13** is an elevation view of one of many embodiments of a stop according to the disclosure.

[0024] FIG. **14** is a perspective view of the stop of FIG. **13**.

[0025] FIG. **15** is an elevation view of another one of many embodiments of a stop according to the disclosure.

[0026] FIG. **16** is a cross sectional view illustrating one of many embodiments of an assembly method according to the disclosure, prior to the poppet being pressed onto the plunger.

[0027] FIG. **17** is a cross sectional view illustrating another of many embodiments of an assembly method according to the disclosure, showing the poppet pressed onto the plunger.

[0028] FIG. **18** is a cross sectional view illustrating yet another of many embodiments of an assembly method according to the disclosure, prior to the cap being pressed into the valve body.

[0029] FIG. **19** is a cross sectional view illustrating still another of many embodiments of an assembly method according to the disclosure, showing the cap pressed into the valve body.

[0030] FIG. **20** is a pneumatic diagram of one of many embodiments of an assembly system according to the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0031] The figures described above and the written description of specific structures and functions below are not presented to limit the scope of what Applicants have invented or the scope of the appended claims. Rather, the figures and written description are provided to teach any person skilled in the art to make and use the inventions for which patent protection is sought. Those skilled in the art will appreciate that not all features of a commercial embodiment of the inventions are described or shown for the sake of clarity and understanding. Persons of skill in this art will also appreciate that the development of an actual commercial embodiment incorporating aspects of the present inventions will require numerous implementation-specific decisions to achieve the developer's ultimate goal for the commercial embodiment. Such implementation-specific decisions may include, and likely are not limited to, compliance with system-related, business-related, government-related and other constraints, which may vary by specific implementation, location and from time to time. While a developer's efforts might be complex and time-consuming in an absolute sense, such efforts would be, nevertheless, a routine undertaking for those of skill in this art having benefit of this disclosure. It must be understood that the inventions disclosed and taught herein are susceptible to numerous and various modifications and alternative forms.

[0032] The use of a singular term, such as, but not limited to, "a," is not intended as limiting of the number of items. Also, the use of relational terms, such as, but not limited to, "top," "bottom," "left," "right," "upper," "lower," "down," "up," "side," and the like are used in the written description for clarity in specific reference to the figures and are not intended to limit the scope of the inventions or the appended claims. The terms "including" and "such as" are illustrative and not limitative. The terms "couple," "coupled," "coupling," "coupler," and like terms are used broadly herein and can include any method or device for securing, binding, bonding, fastening, attaching, joining, inserting therein, forming thereon or therein, communicating, or otherwise associating, for example, mechanically, magnetically, electrically, chemically, operably, directly or indirectly with intermediate elements, one or more pieces of members together and can further include without limitation integrally forming one functional member with another in a unity fashion. The coupling can occur in any direction, including rotationally. Further, all parts and components of the disclosure that are capable of being physically embodied inherently include imaginary and real characteristics regardless of whether such characteristics are expressly described herein, including but not limited to characteristics such as axes, ends, inner and outer surfaces, interior spaces, tops, bottoms, sides, boundaries, dimensions (e.g., height, length, width, thickness), mass, weight, volume and density, among others.

[0033] Any process flowcharts discussed herein illustrate the operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present disclosure. In this regard, each block in a flowchart may represent a module, segment, or portion of code, which can comprise one or more executable instructions for implementing the specified logical function (s). It should also be noted that, in some implementations, the function(s) noted in the block(s) might occur out of the order depicted in the figures. For example, blocks shown in succession may, in fact, be executed substantially concurrently. It will also be noted that each block of flowchart illustration can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0034] Applicants have created new and useful devices, systems and methods for solenoid valves. By press-fitting a poppet onto a plunger, press-fitting a cap into a valve body, press-fitting a stop into a valve body, or any combination thereof, manufacturing of the valve can be simplified, parts can be eliminated, costs can be reduced, operational adjustments can be made, and/or stacked tolerances can be reduced or eliminated.

[0035] FIG. **1** is a cross sectional view of one of many embodiments of a solenoid valve according to the disclosure. FIG. **2** is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure. FIG. **3** is a cross sectional view of a portion of another one of many embodiments of a solenoid valve according to the disclosure. FIG. **4** is a cross sectional perspective view of one of many embodiments of a poppet according to the disclosure. FIG. **5** is a cross sectional view of a portion of yet another one of many embodiments of a solenoid valve according to the disclosure. FIG. **6** is a cross sectional perspective view of another one of many embodiments of a poppet according to the disclosure. FIG. **7** is a cross sectional perspective view of a portion of the poppet of FIG. **6**. FIG. **8** is a cross sectional view of a portion of still another of many embodiments of a solenoid valve according to the disclosure. FIG. **9** is an exploded cross sectional view of a portion of one of many embodiments of a poppet according to the disclosure. FIG. **10** is a cross sectional view of a portion of one of many embodiments of a solenoid valve according to the disclosure. FIG. **11** is a cross sectional view of another portion of one of many embodiments of a solenoid valve according to the disclosure. FIG. **12** is a perspective view of one of many embodiments of a cap according to the disclosure. FIG. **13** is an elevation view of one of many embodiments of a stop according to the disclosure. FIG. **14** is a perspective view of the stop of FIG. **13**. FIG. **15** is an elevation view of another one of many embodiments of a stop according to the disclosure. FIG. **16** is a cross sectional view illustrating one of many embodiments of an assembly method according to the disclosure, prior to the poppet being pressed onto the plunger. FIG. **17** is a cross sectional view illustrating another of many embodiments of an assembly method according to the disclosure, showing the poppet pressed onto the plunger. FIG. **18** is a cross sectional view illustrating yet another of many embodiments of an assembly method according to the disclosure, prior to the cap being pressed into the valve body. FIG. **19** is a cross sectional view illustrating still another of many embodiments of an assembly method according to the disclosure, showing the cap pressed into the valve body. FIG. **20** is a pneumatic diagram of one of many embodiments of an assembly system according to the disclosure. FIGS. **1-20** are described in conjunction with one another.

[0036] In at least one embodiment, a valve **100** according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies **200**, one or more plungers **300** disposed at least partially within the valve body **200**, one or more poppets **400** coupled to the plunger **300**, one or more end caps **500** coupled at least partially within the valve body **200**, one or more stops **600** coupled at least partially within the valve body **200**, or any combination thereof. In at least one embodiment, a valve **100** can include a shell **150** for housing or protecting one or more other components of the valve. In at least one embodiment, the poppet **400** can be press-fit onto the plunger **300**. In at least one embodiment, the cap **500** and/or the stop **600** can limit movement of the plunger **300** and/or poppet **400**. In at least one embodiment, the cap **500** and/or the stop **600** can be press-fit into the valve body **200**.

[0037] In at least one embodiment, the valve body **200** can include one or more port sections **210**, one or more flux collar support sections **220**, one or more coil support sections **230**, or any combination thereof, extending along one or more axes, such as a central longitudinal axis X. In at least one embodiment, the port section **210** can include one or more common ports **212**, one or more normally open ports **214**, and one or more normally closed ports **216**, or any combination thereof. In at least one embodiment, the flux collar support section **220** can support a flux collar, such as to improve efficiency of the valve **100**. In at least one embodiment, the port section **210**, the flux collar support section **220**, and the coil support section **230** of the valve body **200** can be a unitary or monolithic structure, such as by way of being portions of an integrally formed structure (e.g., an injection molded structure).

[0038] In at least one embodiment, the plunger **300** can extend at least partially through the flux collar support section **220** and the coil support section **230**. In at least one embodiment, the plunger **300** can extend at least partially through a flux collar supported by the flux collar support section **220**. In at least one embodiment, the plunger **300** can include one or more stems **310** that extend into the port section **210**. In at least one embodiment, the plunger **300** can include one or more poppets **400** mounted to the stem **310**. In at least one embodiment, one or more biasing members **330**, such as a spring, can bias the plunger **300** and/or poppet **400** towards or away from one or more caps **500**.

[0039] In at least one embodiment, the coil support section **230** of the valve body **200** can include one or more annular walls **232** and/or one or more coils **234**. In at least one embodiment, the wall **232** can support coil **234**, which can selectively actuate, or move, the plunger **300** against the force of the spring **330**, to thereby switch the valve **100** from a normally open position to a normally closed position. In at least one embodiment, the coil support section **230** can include one or more stops **600**, such as a core and/or related structure, to limit the movement of the plunger **300** in one or more directions within the bore **242**. In at least one embodiment, the stop **600** can be a single, unitary structure. In at least one embodiment, the stop **600** can be or include a plurality of structures.

[0040] In at least one embodiment, a valve **100** according to the disclosure can include one or more a poppets **400** press-fit onto the plunger **300** and configured to selectively open and close at least one flow path through the port section **210** of the valve body **200**. In at least one embodiment, the plunger **300** can include a stem **310** extending into the port section **210** of the valve body **200**. In at least one embodiment, the poppet **400** can include one or more shoulders **410** against which the spring **330** can rest and thereby bias the poppet **400** towards a normal operating position, opening a normally open flow path and/or closing a normally closed flow path through the valve body **210**. In at least one embodiment, the coil **234** can overcome the spring **330** to move the plunger **300** and poppet **400** to close a normally open flow path and/or open a normally closed flow path through the valve body **210**.

[0041] In at least one embodiment, the poppet **400** can be press-fit onto the stem **310**. In at least one embodiment, the stem **310** can include one or more annular barbs **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the stem **310** can include one annular barb **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the stem **310** can include a plurality of annular barbs **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**.

[0042] In at least one embodiment, the poppet **400** can include one or more annular grooves **420** configured to engage the barbs(s) **320** on the stem **310** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the poppet **400** can include one annular groove **420** configured to engage the barbs(s) **320** on the stem **310** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the poppet **400** can include a plurality of annular grooves **420** configured to engage the barbs(s) **320** on the stem **310** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, the annular barb(s) **320** and/or the annular groove(s) **420** allow a position of the poppet **400** with respect to the stem **310** to be adjusted, such that the poppet **400** can be mounted along a range of positions along the stem **310**.

[0043] In at least one embodiment, the poppet **400** can be unitary or made of multiple portions. In at least one embodiment, the poppet **400** can include a more rigid portion **430** configured to engage the barb **320** on the stem **310**. In at least one embodiment, the poppet **400** can include a more a resilient portion configured to selectively engage one or more valve seats **240**, **540** to selectively open and close the flow path through the port section **210** of the valve body **200**. In at least one embodiment, with the rigid portion **430** being harder, or more rigid, than the resilient portion **440**, the rigid portion **430** can better hold the poppet **400** in place along the stem **310** of the plunger **300**. In at least one embodiment, with the resilient portion **440** being more resilient, softer, and/or more elastomeric than the rigid portion **430**, the resilient portion **440** can better seal with the valve seats **240**, **540**. In at least one embodiment, the resilient portion **440** can be pulled at least partially into, thru, or onto the rigid portion **430**. In at least one embodiment, the rigid portion **430** can include one or more annular flanges **432**. In at least one embodiment, the resilient portion **440** can include one or more slots **442** configured to mate with the flange **432**.

[0044] In at least one embodiment, the poppet **400** can include one or more bands **450** around a perimeter of the poppet **400** to secure the poppet **400** to the stem **310**. In at least one embodiment, the poppet **400** can be pulled at least partially into, or thru, the band **450**. In at least one embodiment, the band **450** can be metal and/or crimped onto the poppet **400**, thereby securing the poppet **400** to the stem **310**. In at least one embodiment, the band **450** can be a more rigid polymer than the poppet **400**, or the resilient portion **440** thereof, thereby supporting and/or securing the poppet **400** to the stem **310**.

[0045] In at least one embodiment, the poppet **400** can selectively engage one or more valve seats **240**, **540** to selectively open and close one or more flow paths through the port section **210** of the valve body **200**. In at least one embodiment, one or more valve seats **240** can be integral with the valve body **200** and/or can be part of a normally open or normally closed flow path. In at least one embodiment, one or more valve seats **540** can be integral with a cap **500** press-fit into one end of the valve body **200**.

[0046] In at least one embodiment, the poppet **400** can include one or more seals **460**, such as gasket(s) or O-ring (s), configured to selectively engage one or more valve seats **240**, **540** to selectively open and close one or more flow path through the port section **210** of the valve body **200**. In at least one embodiment, the poppet **400** can include a first seal **460** configured to selectively engage a first valve seat **540** to selectively open and close a normally closed flow path through the port section **210** of the valve body **200**. In at least one embodiment, the poppet **400** can include a second seal **460** configured to selectively engage a second valve seat **240** to selectively open and close a normally open flow path through the port section **210** of the valve body **200**. In at least one embodiment, the seal(s) **460** can be mounted to the poppet **400** and/or the stem **310**, with the poppet **400** selectively pressing the seals(s) **460** into the valve seats **240**, **540**.

[0047] In at least one embodiment, portions of the valve body **200** and/or the cap **500** can be resilient. In at least one embodiment, all, or portions, of the poppet **400** can be rigid. In at least one embodiment, the poppet **400** can be rigid and configured to selectively engage a resilient valve seat **240**, **540** to selectively open and close the flow path through the port section **210** of the valve body **200**.

[0048] In at least one embodiment, a valve **100** according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies **200** having a port section **210** and a coil support section **230** extending along a longitudinal axis, one or more plungers **300** within the valve body **200** and extending at least partially through the coil support section **230**, a poppet **400** including a resilient portion **440** configured to selectively engage a rigid valve seat **240**, **540** to open and close at least one flow path through the port section **210** of the valve body **200**, or any combination thereof. In at least one embodiment, the plunger **300** can include one or more stems **310** extending into the port section **210** of the valve body **200**. In at least one embodiment, the stem **310** can include at least one annular barb **320** configured to engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**. In at least one embodiment, one or more of the valve seats **240** can be integral with the valve body **200**. In at least one embodiment, one or more of the valve seats **540** can be integral with a cap **500** press-fit into one end of the valve body **200**.

[0049] In at least one embodiment, the poppet **400** can include one or more rigid portions **430** configured to engage the barb **320** on the stem **310**. In at least one embodiment, the rigid portion **430** can be harder than the resilient portion **440**. In at least one embodiment, the rigid portion **430** can include one or more an annular flanges **432**. In at least one embodiment, the resilient portion **440** can be configured to mate with the flange **432** and/or can include one or more slots **442** configured to mate with the flange **432**.

[0050] In at least one embodiment, a valve **100** according to the disclosure, such as a miniature solenoid valve, can include one or more valve bodies **200** having a port section **210** and a coil support section **230** extending along a longitudinal axis, one or more plungers **300** within the valve body **200** and extending at least partially through the coil support section **230**, one or more poppets **400** configured to selectively engage one or more valve seats **240**, **540** to open and close one or more flow path through the port section **210** of the valve body **200**, or any combination thereof. In at least one embodiment, the plunger **300** can include a stem **310** extending into the port section **210** of the valve body **200**. In at least one embodiment, the stem **310** can include one or more annular barbs **320**. In at least one embodiment, the barbs **320** can engage an interior of the poppet **400** to hold the poppet **400** in place on the stem **310**.

[0051] By press-fitting the poppet **400** onto the stem **310** of the plunger **300**, manufacturing of the valve **100** can be simplified, parts can be eliminated, costs can be reduced, operational adjustments can be made, and/or stacked tolerances can be reduced or eliminated. For example, a poppet body or holder can be eliminated. In at least one embodiment, the poppet **400** according to the disclosure does not require a complex and/or expensive over molding process. In at least one embodiment, the poppet **400** according to the disclosure can include a myriad of techniques, as described above, to provide adequate holding force, or resistance to pull-out, to remain in place on the plunger **300** for proper functionality within the valve **100**. In at least one embodiment, the poppet **400** according to the disclosure can be a monolithic or unitary elastomeric, or otherwise resilient, structure with integral shoulder **410** for the spring **330** to provide a bias force against in order to operate the valve **100**. In at least one embodiment, one or more of the barbs **320** and/or grooves **420** can be spiral, thereby allowing the poppet **400** according to the disclosure to be press-fit and/or threaded onto the stem **310**, while still achieving many or all of the advantages discussed herein.

[0052] In at least one embodiment, an external surface of the stem **310** and/or an internal surface of the poppet **400** can be rough, or roughened, to increase static friction between the stem **310** and the poppet **400**. In at least one embodiment, a dissolvable and/or evaporable lubricant can be used during assembly of the valve **100**, such as to temporarily reduce friction between the stem **310** and the poppet **400**. In at least one embodiment, an adhesive can be used to bond the stem **310** and the poppet **400**. In at least one embodiment, the poppet **400** can include a through hole in order to inject an adhesive between the stem **310** and the poppet **400**. In at least one embodiment, the poppet **400** can be designed to allow for transfer of heat or ultraviolet light to the adhesive between the stem **310** and the poppet **400**. In at least one embodiment, the poppet **400** according to the disclosure can be vulcanized, hardened, shrunk, or any combination thereof once in place on the stem **310**. In at least one embodiment, the poppet **400** according to the disclosure, or a portion thereof, can be metal and/or a rigid polymer and/or the valve seats **240**, **540** can be metal and/or a rigid polymer, with the valve **100** relying on a metal-to-metal and/or metal-to-plastic seal, thereby eliminating elastomeric materials in sealing the flow path(s).

[0053] In at least one embodiment, a valve **100** according to the disclosure can include one or more caps **500** press-fit at least partially into the valve body **200** at an end of the valve body **200**, such as at the port section **210** of the valve body. In at least one embodiment, the cap **500** can include one or more valve seats **540** that can be selectively engaged by the poppet **400** mounted to the plunger **300** to selectively open and close a flow path through the port section **210** of the valve body **200**. In at least one embodiment, the cap **500** can include one or more annular barbs **520** that sealingly mate with the valve body **200**. In at least one embodiment, the barb **520** can extend around a perimeter of the cap **500**. In at least one embodiment, the barb **520** can resist pressure within the valve body **200** to hold the cap **500** in place within or relative to the valve body **500**. In at least one embodiment, the annular barb(s) **520** can allow a position of the cap **500** within the valve body **200** to be adjusted, such that the cap **500** can be mounted along a range of positions along the valve body **200**. Adjusting the position of the cap **500** within or relative to the valve body **200** can adjust the stroke of the plunger **300**, one or more flow rates through the port section **210** of the valve body, reduce stacked tolerances in the valve **100**, or any combination thereof.

[0054] In at least one embodiment, the barb **520** can be integral with the cap **500** and/or can hermetically or fluidically seal the cap **500** to the valve body **200**, which can include doing so without any additional sealing device, such as an O-ring, gasket, or sealant. In at least one embodiment, the barb **520** and/or the cap **500** can be metal and/or the valve body **200** can be a resilient material. In at least one embodiment, the barb **520** can deform the valve body **200**, such as while being press-fit therein, thereby sealing the cap **500** to the valve body **200**.

[0055] In at least one embodiment, the cap **500** can include one annular barb **520** that extends around a perimeter of the cap **500**. In at least one embodiment, the cap **500** can include two or more annular barbs **520** that extend around a perimeter of the cap **500**. In at least one embodiment, the barbs **520** can resist pressure within the valve body **200** to hold the cap **500** in place within or relative to the valve body **200**.

[0056] In at least one embodiment, a valve **100** according to the disclosure can include one or more stops **600** press-fit at least partially into the valve body **200** at an end of the valve body **200**, such as at the coil section **230** of the valve body **200**. In at least one embodiment, the stop **600** can include one or more shafts **610** to limit movement of the plunger **300** away from (i.e., in a direction away from) the cap **500** and/or a flange **612**, which can fix a position of the stop **600** with respect to the valve body **200**. In at least one embodiment, the stop **600** can include one or more annular barbs **620** that sealingly mates with the valve body **200**. In at least one embodiment, the barb **620** can extend around a perimeter of the stop **600**. In at least one embodiment, the barb **620** can extend around a perimeter of the shaft **610** and/or the flange **612**. In at least one embodiment, the barb **620** can resist pressure within the valve body **200** to hold the stop **600** in place within or relative to the valve body **200**. In at least one embodiment, the annular barb(s) **620** can allow a position of the stop **600** within or relative to the valve body **200** to be adjusted, such that the stop **600** can be mounted along a range of positions along the valve body **200**. Adjusting the position of the stop **600** within or relative to the valve body **200** can adjust the stroke of the plunger **300**, and/or reduce stacked tolerances in the valve **100**.

[0057] In at least one embodiment, the barb **620** can be integral with the stop **600** and/or can hermetically or fluidically seal the stop **600** to the valve body **200**, which can include doing so singlehandedly, such as without the aid of an additional sealing device, such as an O-ring, gasket, or sealant. In at least one embodiment, the barb **620** and/or the stop **600** can be metal and/or the valve body **200** can be a resilient material. In at least one embodiment, the barb **620** can be configured to deform the valve body **200**, such as while being press-fit therein, thereby sealing the stop **600** to the valve body **200**.

[0058] In at least one embodiment, the stop **600** can include one or more annular barbs **620** that extend around a perimeter of the stop **600**. In at least one embodiment, the stop **600** can include two or more annular barbs **620** that extend around a perimeter of the stop **600**. In at least one embodiment, the barbs **620** can resist pressure within the valve body **200** and/or can hold the stop **600** in place within or relative to the valve body **200**.

[0059] In at least one embodiment, a valve **100** according to the disclosure can include one or more valve bodies **200** having a port section **210** and a coil support section **230** extending along a longitudinal axis, a plunger **300** within the valve body **200** and extending at least partially through the coil support section **230**, a cap **500** press-fit at least partially into the valve body **200** at a first end of the valve body **200**, a stop **600** press-fit at least partially into the valve body **200** at a second, opposite end of the valve body **200**, or any combination thereof. In at least one embodiment, the cap **500** can include a valve seat **540** configured to be selectively engaged by a poppet **400** mounted to the plunger **300** to selectively open and close a flow path through the port section **210** of the valve body **200**. In at least one embodiment, the cap **500** can include one or more first annular barbs **520** extending around a perimeter of the cap **500**, which can sealingly secure the cap **500** at least partially within the valve body **200**. In at least one embodiment, the stop **600** can limit movement of the plunger **300** away from the cap **500**. In at least one embodiment, the stop **600** can include one or more second or other annular barbs **620** extending around a perimeter of the stop **600**, which can sealingly secure the stop **600** at least partially within the valve body **200**.

[0060] In at least one embodiment, the first barb **520** can resist pressure within the valve body **200** to hold the cap **500** in place within or relative to the valve body **200**. In at least one embodiment, the first barb **520** can be metal. In at least one embodiment, the first barb **520** and the cap **520** can be integrally formed of metal. In at least one embodiment, the valve body **200** can be a resilient material. In at least one embodiment, the first barb **520** can deform the valve body **200**, such as while being press-fit therein, thereby sealing the cap **500** to the valve body **200**.

[0061] In at least one embodiment, the second barb **620** can resist pressure within the valve body **200** to hold the stop **600** in place within or relative to the valve body **200**. In at least one embodiment, the barb can be metal. In at least one embodiment, the second barb and the stop can be integrally formed of metal. In at least one embodiment, the valve body can be a resilient material. In at least one embodiment, the second barb can deform the valve body, such as while being press-fit therein, thereby sealing the stop **600** to the valve body.

[0062] Utilizing barbs **520**, **620** on the cap **500** and/or stop **600**, any of which can be metallic, advantageously provides retention and/or hermetic sealing within the valve body **200**, which can be plastic or polymer, that requires no additional parts to function, reduces the number of parts required, reduces time and/or effort required for assembly, allows for positional fixation and/or adjustment within the valve body **200**, or any combination thereof. In at least one embodiment, the cap **500** can include two or more barbs **520** to prevent angular misalignment of the valve seat **540** to the valve body **200** and/or plunger **400**. In at least one embodiment, heat can be used to shrink, stake, and/or further couple or seal the valve body **200** to the cap **500** and/or the stop **600**.

[0063] In at least one embodiment, a valve **100** according to the disclosure can include one or more valve bodies **200** having a port section **210** and a coil support section **230** extending along a longitudinal axis, a plunger **300** within the valve body **200** and extending at least partially through the coil support section **230**, a poppet **400** press-fit onto the plunger **300**, or any combination thereof. In at least one embodiment, the body **200** can include one or more first valve seats **240**. In at least one embodiment, the plunger **300** can include a stem **310** extending through the first valve **240** seat into the port section **210**. In at least one embodiment, the poppet **400** can be press-fit onto the stem **310** of the plunger **400**. In at least one embodiment, the poppet **400** can selectively engage the first valve seat **240** to open and close a first flow path through the port section **210** of the valve body **200**. In at least one embodiment, a stroke of the plunger **300** and/or a flow rate through the first flow path can be adjusted by adjusting a first position of the poppet **400** on the plunger **300**.

[0064] In at least one embodiment, a valve **100** according to the disclosure can include one or more caps **500** press-fit at least partially into the port section **210** of the valve body **200**. In at least one embodiment, the cap **500** can include one or more second valve seat **540**. In at least one embodiment, the poppet **400** can selectively engage the second valve seat **540** to open and close a second flow path through the port section **210** of the valve body **200**. In at least one embodiment, a stroke of the plunger **300** and/or a flow rate through the second flow path can be adjusted by adjusting a second position of the cap **500** in or relative to the body **200**.

[0065] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a first flow between a first port and a second port of the valve **100** and/or pressing a poppet **400** onto a plunger **300** of the valve **100** until the first flow stops. In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a second fluid flow between the first port and a third port of the valve **100**, energizing a coil **234** of the valve **100**, pressing a cap **500** into a body **200** of the valve **100** until the second fluid flow stops, or any combination thereof. In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a second fluid flow between the first port and a third port of the valve **100** and/or pressing a cap **500** into a body **200** of the valve **100** until a desired rate of the second fluid flow is achieved. In at least one embodiment, the first port can be a common port. In at least one embodiment, the second port can be a normally closed port. In at least one embodiment, the third port can be a normally open port.

[0066] In at least one embodiment, an assembly system or apparatus **700**, such as that shown in FIG. **20** for exemplary purposes, can be used to provide a controlled fluid flow to/from one or more ports of the port section **210** of the valve body **200** during assembly of valve **100**. In at least one embodiment, such fluid flow can be of air, another inert gas, water, another inert fluid, or any combination thereof. In at least one embodiment, the system **700** can be or include a pneumatic system, one or more assembly fixtures (not shown) and one or more presses or pressing machines (not shown), such as a dynamic precision press capable of precision pressing of components based on closed loop sensor feedback. In at least one embodiment, the system **700** can one or more supply pressure inlets, one or more solenoid valves (e.g., solenoid valves SV1, SV2, et seq.), one or more pressure regulators (e.g., pressure regulators PR1, PR2), one or more flow meters (e.g., flow meter FM), one or more level meters (e.g., level meter LM), one or more pressure transmitters (e.g., pressure transmitter PT), one or more orifices, one or more valve port connections, one or more vents to atmosphere, or any combination thereof.

[0067] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a power consumption of the coil **234** and continuing to press the cap **500** into the body **200** and/or the poppet **400** onto the plunger **300**, until the power consumption matches a desired setpoint. By adjusting the position of the cap **500** in or relative to the body **200** and/or the poppet **400** on the plunger **300**, power consumption of the coil **234** can be controlled.

[0068] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include energizing a coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, and monitoring for a sound caused by a plunger **300** of the valve **100** contacting a stop **600** of the valve **100**. In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include repeating the steps of pressing the poppet **400** onto the plunger **300**, energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for the sound caused by the plunger **300** contacting the stop **600**, or any combination thereof, until the sound caused by the plunger **300** contacting the stop **600** is no longer detected. By adjusting the position of the poppet **400** on the plunger **300**, the stroke of the plunger **300** can be controlled, such as to limit the engagement between the plunger **300** and the stop **600**.

[0069] In at least one embodiment, a method of assembling a valve **100** according to the disclosure can include monitoring a first flow between a common port and a normally closed port of the valve **100**, pressing a poppet **400** onto a plunger **300** of the valve **100** until the first flow stops, monitoring a second fluid flow between the common port and a normally open port of the valve **100**, energizing a coil **234** of the valve **100**, pressing a cap **500** into a body **200** of the valve **100** until the second fluid flow stops, or any combination thereof. In at least one embodiment, the method of assembling a valve **100** according to the disclosure can further include monitoring a power consumption of the coil **234** and/or continuing to press, or repeatedly pressing, the cap **500** into the body **200** and/or the poppet **400** onto the plunger **300**, until the power consumption matches a desired setpoint. In at least one embodiment, the method of assembling a valve **100** according to the disclosure can further include energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for a sound caused by a plunger **300** of the valve **100** contacting a stop **600** of the valve **100**, or any combination thereof. In at least one embodiment, the method of assembling a valve **100** according to the disclosure can further include repeating pressing the poppet **400** onto the plunger **300**, energizing the coil **234** of the valve **100**, de-energizing the coil **234** of the valve **100**, monitoring for the sound caused by the plunger **300** contacting the stop **600**, or any combination thereof, until the sound caused by the plunger **300** contacting the stop **600** is no longer detected. In at least one embodiment, the poppet **400** can be pressed onto the plunger **300** before the cap **500** is pressed into the body **200**.

[0070] By monitoring flow rates through the port section **210** of the valve **100** while pressing the poppet **400** onto the plunger **300** and/or the cap **500** into the body **200** the position of the poppet **400** on the plunger **300** and/or the cap **500** in the body **200** can be adjusted, thereby reducing, eliminating, or otherwise mitigating the effects of stack-up tolerances in the valve **100**, such as those impacting the stroke of the plunger **300**. Pressing the poppet **400** further onto the plunger **300** and/or pressing the cap **500** further into the body **200** can adjust the stroke of the plunger **300**, flow through the port section of the valve **100**, power consumption of the coil **234**, operational noise within the valve **100**, load and/or preload on the spring **330**, or any combination thereof. The assembly method described herein can reduce costs through increased tolerances (i.e., the adjustability reduces the effect of tolerances, thereby allowing for looser tolerances in individual component manufacturing) provide better performance by reducing excess elastomer deformation on the valve seats **240**, **540**, mitigate against energized leak by conducting a pressing operation until enough elastomer deformation has occurred to adequately seal between the cap **500** and the body **200** and/or between the stop **600** and the body **200**, offer precise flow control, offer variability in flow rates without change to orifice size, increase reliability of the valve **100** (e.g., since functionality can be confirmed during assembly), or any combination thereof. Stack-up tolerances can be mitigated through the use of the annular barbs **320**, **520**, **620** described herein, threaded barbs, parts sorting (such as to counter tolerances), or any combination thereof.

[0071] In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet press-fit onto the plunger and configured to selectively open and close at least one flow path through the port section of the valve body, or any combination thereof. In at least one embodiment, the plunger can include a stem extending into the port section of the valve body. In at least one embodiment, the poppet can be press-fit onto the stem. In at least one embodiment, the stem can include one or more annular barbs to engage an interior of the poppet to hold the poppet in place on the stem. In at least one embodiment, the poppet can be configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body.

[0072] In at least one embodiment, the poppet can be rigid and/or configured to selectively engage a resilient valve seat to selectively open and close the flow path through the port section of the valve body. In at least one embodiment, the poppet can include a rigid portion configured to engage the barb on the stem and/or a resilient portion configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body. In at least one embodiment, the rigid portion can be harder than the resilient portion. In at least one embodiment, the rigid portion can include an annular flange. In at least one embodiment, the resilient portion can mate with the flange.

[0073] In at least one embodiment, the poppet can include a band around a perimeter of the poppet to secure the poppet to the stem. In at least one embodiment, the band can be crimped onto the poppet, thereby securing the poppet to the stem.

[0074] In at least one embodiment, the valve seat can be integral with the valve body. In at least one embodiment, the valve seat can be integral with a cap press-fit into one end of the valve body. In at least one embodiment, the poppet can include a seal configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body. In at least one embodiment, the poppet can include a first seal configured to selectively engage a first valve seat to selectively open and close a normally closed flow path through the port section of the valve body. In at least one embodiment, the poppet can include a second seal configured to selectively engage a second valve seat to selectively open and close a normally open flow path through the port section of the valve body.

[0075] In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet including a resilient portion configured to selectively engage a rigid valve seat to open and close at least one flow path through the port section of the valve body, or any combination thereof. In at least one embodiment, the plunger can include a stem extending into the port section of the valve body. In at least one embodiment, the stem can include at least one annular barb configured to engage an interior of the poppet to hold the poppet in place on the stem. In at least one embodiment, the valve seat can be integral with the valve body. In at least one embodiment, the valve seat can be integral with a cap press-fit into one end of the valve body.

[0076] In at least one embodiment, the poppet can include a rigid portion configured to engage the barb on the stem. In at least one embodiment, the rigid portion can be harder than the resilient portion. In at least one embodiment, the rigid portion can include an annular flange. In at least one embodiment, the resilient portion can mate with the flange.

[0077] In at least one embodiment, a valve can include a valve body having a port section and a coil support section extending along a longitudinal axis, a plunger within the valve body and extending at least partially through the coil support section, a poppet configured to selectively engage a valve seat to open and close a flow path through the port section of the valve body, or any combination thereof. In at least one embodiment, the plunger can include a stem extending into the port section of the valve body. In at least one embodiment, the stem can include a plurality of annular barbs. In at least one embodiment, the barbs can engage an interior of the poppet to hold the poppet in place on the stem.

[0078] Other and further embodiments utilizing one or more aspects of the disclosure can be devised without departing from the spirit of Applicants' disclosure. For example, the devices, systems and methods can be implemented for numerous different types and sizes in numerous different industries. Further, the various methods and embodiments of the devices, systems and methods can be included in combination with each other to produce variations of the disclosed methods and embodiments. Discussion of singular elements can include plural elements and vice versa. The order of steps can occur in a variety of sequences unless otherwise specifically limited. The various steps described herein can be combined with other steps, interlineated with the stated steps, and/or split into multiple steps. Similarly, elements have been described functionally and can be embodied as separate components or can be combined into components having multiple functions.

[0079] The inventions have been described in the context of preferred and other embodiments and not every embodiment of the inventions has been described. Obvious modifications and alterations to the described embodiments are available to those of ordinary skill in the art having the benefits of the present disclosure. The disclosed and undisclosed embodiments are not intended to limit or restrict the scope or applicability of the inventions conceived of by the Applicants, but rather, in conformity with the patent laws, Applicants intend to fully protect all such modifications and improvements that come within the scope or range of equivalents of the following claims.

What is claimed is:

1. A valve comprising:

a valve body having a port section and a coil support section extending along a longitudinal axis;

a plunger within the valve body and extending at least partially through the coil support section; and

a poppet press-fit onto the plunger and configured to selectively open and close at least one flow path through the port section of the valve body.

2. The valve of claim **1**, wherein the plunger includes a stem extending into the port section of the valve body; and wherein the poppet is press-fit onto the stem.

3. The valve of claim **2**, wherein the stem includes an annular barb configured to engage an interior of the poppet to hold the poppet in place on the stem.

4. The valve of claim **2**, wherein the stem includes a plurality of annular barbs configured to engage an interior of the poppet to hold the poppet in place on the stem.

5. The valve of claim **4**, wherein the poppet includes a rigid portion configured to engage the barb on the stem and a resilient portion configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body; and wherein the rigid portion is harder than the resilient portion.

6. The valve of claim **5**, wherein the rigid portion includes an annular flange; and wherein the resilient portion is configured to mate with the flange.

7. The valve of claim **2**, further comprising a band around a perimeter of the poppet to secure the poppet to the stem.

8. The valve of claim **7**, wherein the band is crimped onto the poppet, thereby securing the poppet to the stem.

9. The valve of claim **1**, wherein the poppet is configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body.

10. The valve of claim **9**, wherein the valve seat is integral with the valve body.

11. The valve of claim **9**, wherein the valve seat is integral with a cap press-fit into one end of the valve body.

12. The valve of claim **1**, wherein the poppet includes a seal configured to selectively engage a valve seat to selectively open and close the flow path through the port section of the valve body.

13. The valve of claim **1**, wherein the poppet includes a first seal configured to selectively engage a first valve seat to selectively open and close a normally closed flow path through the port section of the valve body; and wherein the poppet includes a second seal configured to selectively engage a second valve seat to selectively open and close a normally open flow path through the port section of the valve body.

14. The valve of claim **1**, wherein the poppet is rigid and is configured to selectively engage a resilient valve seat to selectively open and close the flow path through the port section of the valve body.

15. A valve comprising:

a valve body having a port section and a coil support section extending along a longitudinal axis;

a plunger within the valve body and extending at least partially through the coil support section, wherein the plunger includes a stem extending into the port section of the valve body; and a poppet including a resilient portion configured to selectively engage a rigid valve seat to open and close at least one flow path through the port section of the valve body;

wherein the stem includes at least one annular barb configured to engage an interior of the poppet to hold the poppet in place on the stem.

16. The valve of claim **15**, wherein the poppet comprises a rigid portion configured to engage the barb on the stem; and wherein the rigid portion is harder than the resilient portion.

17. The valve of claim **16**, wherein the rigid portion includes an annular flange; and wherein the resilient portion is configured to mate with the flange.

18. The valve of claim **15**, wherein the valve seat is integral with the valve body.

19. The valve of claim **15**, wherein the valve seat is integral with a cap that is press-fit at least partially into one end of the valve body.

20. A valve comprising:

a valve body having a port section and a coil support section extending along a longitudinal axis;

a plunger within the valve body and extending at least partially through the coil support section, wherein the plunger includes a stem extending into the port section of the valve body, and wherein the stem includes a plurality of annular barbs; and

a poppet configured to selectively engage a valve seat to open and close a flow path through the port section of the valve body;

wherein the barbs are configured to engage an interior of the poppet to hold the poppet in place on the stem.

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